

Revised Generation Interconnection System Impact Study Report

For Queue Project Q16-0581

Hemdon 138 kV
December 31, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0581. The Interconnection Customer proposes a 302.0 MW wind facility interconnecting at the Hemdon 138 kV substation in the Virginia area. The request was received on February 10, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 213 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$13,458,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

HEMDON – CRYSTAL LAKE 138.0 kV: 124.6% loading (276 MVA) following the outage of the DEL AMO – GRAND BLANC GENERATING STATION 138.0 kV; DFAX = 76.3%.

The ANTHEM – DRANESVILLE 138.0 kV is overloaded to 115.3% of its 242 MVA emergency rating for the loss of the FOSTER DAM – MCCORMICK 138.0 kV. The project contributes 5.1% DFAX (15.4 MW) to this overload.

For the loss of DEL AMO – BULL RUN 138.0 kV line, the HEMDON – LUNENBURG 138.0 kV L loads to 113.5% of its 248 MVA rating. Project DFAX: 23.7%.

For the loss of RUSSELL FORK – ROBISON PARK 230.0 kV, the DEKALB – JONES CHAPEL SUBSTATION 500.0 kV loads to 109.6% of its 2,860 MVA rating. Project DFAX: 11.7%.

The ELK GARDEN – CARDEROCK 138.0 kV is overloaded to 105.8% of its 239 MVA emergency rating for the loss of the HELIJET – ELK GARDEN 138.0 kV line. The project contributes 5.4% DFAX (16.2 MW) to this overload.

The DRANESVILLE – MCCORMICK 138.0 kV Circuit 2 is overloaded to 104.8% of its 326 MVA emergency rating for the loss of the DRANESVILLE – MCCORMICK 138.0 kV. The project contributes 7.5% DFAX (22.7 MW) to this overload.

For the loss of FOSTER DAM – MCCORMICK 138.0 kV, the ELK GARDEN – ANTHEM 138.0 kV loads to 103.6% of its 229 MVA rating. Project DFAX: 5.1%.

DRANESVILLE – MCCORMICK SUBSTATION 138.0 kV Line 1: 102.9% loading (332 MVA) following the outage of the DRANESVILLE – MCCORMICK 138.0 kV Line; DFAX = 7.5%.

For the loss of DEKALB – RICHLAND 500.0 kV Line, the DEKALB 230 kV/1 kV TX loads to 101.7% of its 1,789 MVA rating. Project DFAX: 6.6%.

The DEKALB 500-1 kV Transformer is overloaded to 101.6% of its 1,790 MVA emergency rating for the loss of the RICHLAND – DEKALB 500.0 kV. The project contributes 6.6% DFAX (20.0 MW) to this overload.

DEKALB – RICHLAND 500.0 kV: 101.2% loading (2,568 MVA) following the outage of the STATE ST – ROBISON PARK 230.0 kV; DFAX = 9.9%.

The ROBISON PARK – RUSSELL FORK 230.0 kV is overloaded to 100.6% of its 2,127 MVA emergency rating for the loss of the DEKALB – JONES CHAPEL 500.0 kV. The project contributes 7.8% DFAX (23.6 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
HEMDON – CRYSTAL LAKE 138.0 kV	Rebuild line with higher-capacity conductor	\$2,872,000	40% (\$1,140,000)
ANTHEM – DRANESVILLE 138.0 kV	Rebuild line with higher-capacity conductor	\$4,631,000	89% (\$4,135,000)
HEMDON – LUNENBURG 138.0 kV L	Rebuild line with higher-capacity conductor	\$729,000	100% (\$729,000)
DEKALB – JONES CHAPEL SUBSTATION 500.0 kV	Reconductor line	\$36,278,000	2% (\$672,000)
ELK GARDEN – CARDEROCK 138.0 kV	Reconductor line	\$1,414,000	50% (\$710,000)
DRANESVILLE – MCCORMICK 138.0 kV #2	Reconductor line	\$1,409,000	45% (\$634,000)
ELK GARDEN – ANTHEM 138.0 kV L	Reconductor line	\$2,574,000	89% (\$2,298,000)
MCCORMICK – DRANESVILLE 138.0 kV	Reconductor line	\$1,109,000	45% (\$499,000)
DEKALB 230/1kV Autotransformer	Replace transformer with higher-rated unit	\$9,342,000	2% (\$173,000)
DEKALB 500-1 kV Transformer	Replace transformer with higher-rated unit	\$34,296,000	2% (\$636,000)
RICHLAND – DEKALB 500.0 kV	Reconductor line	\$88,885,000	2% (\$1,647,000)
ROBISON PARK – RUSSELL FORK 230.0 kV	Reconductor line	\$9,973,000	2% (\$185,000)
	Total allocated to project		\$13,458,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.